



Registration-free visible-thermal fusion for UAV detection

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ABSTRACT

Visible-thermal (RGB-T) fusion provides complementary cues for UAV detection, yet most existing methods assume well-aligned cross-modal inputs. This assumption is difficult to satisfy in practical dual-sensor systems, where RGB and thermal infrared images often suffer from spatial displacement, scale variation, and temporal offset. Directly fusing such unregistered image pairs can introduce feature conflicts, especially for small UAVs in cluttered scenes. To address this problem, we propose RFVTD, a registration-free visible-thermal UAV detection framework that operates on unaligned RGB-T image pairs without image-level pre-registration or pixel-level alignment supervision. RFVTD follows a coarse-to-fine cross-modal feature interaction strategy. First, a Shared-Private Decomposition Module (SPDM) separates modality-invariant shared cues from modality-specific private cues to reduce negative transfer. Second, an Adaptive Modality Semantic Association and Enhancement Module (AMSEM) establishes coarse bidirectional semantic associations between RGB and T shared features, aggregates non-local cross-modal cues, derives initial deformation priors, and enhances small-UAV responses with reliability-aware modality weighting. Third, a Dual-Supervised Feature Alignment (DSFA) module refines residual deformation in both RGB-to-T and T-to-RGB directions under dual-modality detection supervision. Experiments on Anti-UAV, DroneVehicle, DVTOD, and LLVIP show that RFVTD achieves strong performance across different alignment conditions, with clear advantages in severely misaligned scenarios and competitive results on well-aligned data. Real-world evaluations further demonstrate its robustness against complex background interference.

1. Introduction

Unmanned aerial vehicles (UAVs) have been widely deployed in urban surveillance [1], logistics [2], and agriculture [3], owing to their low cost, high mobility, and flexible deployment [4]. At the same time, unauthorized UAVs have raised increasing concerns in public safety and security [5,6]. Accurate detection and localization of UAVs are therefore essential for subsequent countermeasures, such as warning, tracking, jamming, or interception [5]. Compared with conventional object detection [7], anti-UAV detection is more challenging [8,9]. UAVs usually occupy only a few pixels at long distances and exhibit weak visual cues. They also appear in complex backgrounds, including sky, forests, and urban scenes, where clouds, birds, and cluttered structures can easily cause false alarms [10,11].

Visible-thermal (RGB-T) fusion provides a promising solution to these challenges [12–14]. RGB images provide rich texture, color, and

edge information, while thermal infrared (T) images capture heat signatures and are more robust under low illumination or nighttime conditions. These complementary properties are particularly useful for detecting small UAVs in adverse environments. However, effective RGB-T fusion usually assumes that the two modalities are spatially aligned or have reliable cross-modal correspondence. This assumption is difficult to satisfy in practical anti-UAV systems.

In real-world UAV monitoring scenarios, RGB-T image pairs are often spatially misaligned due to dual-lens parallax, sensor displacement, mechanical vibration, and temporal offsets [15,16]. Manual registration is labor-intensive and unsuitable for real-time deployment. Standalone automatic registration methods also have inherent limitations, as they typically rely on pixel-level alignment supervision or stable cross-modal textures, both of which are difficult to obtain in UAV-oriented scenarios. More importantly, registration and fusion should not be treated as two isolated stages; instead, they should be jointly optimized to reinforce

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each other [17,18]. Therefore, a simple cascaded pipeline that performs registration before fusion may suffer from accumulated registration errors, inconsistent optimization objectives, and limited task adaptivity. Conversely, directly applying existing RGB–T fusion methods to unregistered inputs may introduce feature conflicts, spatial response confusion, and negative transfer [19]. These issues become even more severe when small targets, large spatial displacements, and modality imbalance occur simultaneously.

Recent studies have explored misalignment-aware visible–thermal detection [20–22]. However, most methods target weak perturbations or rely on local alignment, limiting their effectiveness under large RGB–T displacement and strong modality-specific responses. These limitations arise from two aspects. First, directly aligning mixed cross-modal features may force modality-specific details into a shared space and cause negative transfer. Second, local deformation predictors lack non-local correspondence cues and therefore struggle with long-range displacement. A more reliable strategy is to separate modality-invariant structures from modality-specific details, establish coarse non-local semantic correspondence, and then refine residual deformation in a task-driven manner.

To this end, we propose Registration-Free Visible–Thermal Fusion for UAV Detection (RFVTD), a detection-oriented framework that directly processes unregistered RGB–T image pairs without explicit image-level registration. Here, “registration-free” means that RFVTD does not require image-level pre-registration or pixel-level alignment supervision, while feature-level alignment is learned jointly with detection. RFVTD follows a coarse-to-fine feature interaction strategy. First, the Shared–Private Decomposition Module (SPDM) separates each modality feature into a shared component and a private component. The shared component captures modality-invariant structural cues for cross-modal association and alignment, while the private component preserves modality-specific discriminative information. Second, the Adaptive Modality Semantic Association and Enhancement Module (AMSEM) establishes coarse bidirectional semantic associations between RGB and T shared features. It further aggregates non-local cross-modal cues, derives initial deformation priors, and enhances small-UAV responses with reliability-aware modality weighting. Third, the Dual-Supervised Feature Alignment (DSFA) module refines the residual deformation in both RGB-to-T and T-to-RGB directions under dual-modality detection supervision. In this way, RFVTD avoids relying on pre-registration and reduces the burden of estimating large displacement with purely local predictors.

We evaluate RFVTD on four RGB–T benchmarks, including Anti-UAV [8], DroneVehicle [23], DVTOD [22], and LLVIP [24]. These datasets cover a broad range of alignment conditions, from severely misaligned UAV imagery to well-aligned low-light scenes. Experimental results show that RFVTD is particularly effective under misaligned conditions and remains competitive on aligned data. Ablation studies validate the contribution of SPDM, AMSEM, and DSFA, while additional real-world evaluations further demonstrate the robustness of the proposed framework under complex background interference.

The main contributions are summarized as follows:

- We propose RFVTD, a detection-oriented registration-free visible–thermal framework for UAV detection. It directly handles spatially misaligned RGB–T image pairs without relying on image-level pre-registration or pixel-level alignment supervision.
- We design a coarse-to-fine cross-modal feature interaction scheme. SPDM separates modality-invariant shared cues from modality-specific private cues; AMSEM establishes coarse bidirectional semantic associations and derives initial deformation priors; DSFA further refines residual deformation under dual-modality detection supervision.
- We conduct comprehensive experiments on four RGB–T benchmarks and additional real-world data. The results demonstrate the effective-

ness of RFVTD under challenging misaligned conditions and verify the complementary roles of the proposed modules.

2. Related work

2.1. Visible–thermal object detection

Early visible–thermal detectors typically adopt dual-stream CNNs, with RGB and T features extracted separately and fused at different stages. Halfway Fusion [25] compared several fusion strategies and showed the effectiveness of mid-level fusion on KAIST [26]. Later methods introduced illumination-aware gating [27], attention-based cross-modal interaction [28], and gated fusion units [29,30] to adaptively balance modality contributions and suppress redundant responses. Transformer-based methods, including C²Former [31], MS-DETR [32], Cross-Modality Fusion Transformer [33], and TokenFusion [34], further improve long-range cross-modal interaction. However, most RGB–T detectors still assume well-aligned inputs, which is difficult to guarantee in UAV scenarios because of parallax, sensor displacement, and dynamic sensing conditions. Benchmarks such as DroneVehicle [23] and DVTOD [22] further highlight the need to handle spatially misaligned RGB–T image pairs.

2.2. Anti-UAV detection

Anti-UAV detection is challenging because UAV targets are usually small, fast-moving, and easily confused with environmental distractors such as birds, clouds, and cluttered structures. Recent studies [10,35,36] categorize anti-UAV techniques into radar-based, radio-frequency-based, and vision-based approaches. Among them, vision-based methods [5,37] have attracted increasing attention due to their low cost and intuitive deployment. Large-scale benchmarks such as Anti-UAV [8] and Anti-UAV410 [9] have promoted the development of UAV detection and tracking methods, especially under occlusion, scale variation, motion blur, and complex backgrounds. In addition, the Dronevs-Bird Challenge [11] highlights the difficulty of distinguishing UAVs from visually similar distractors. Multimodal sensing provides complementary cues for anti-UAV detection. RGB images preserve texture and structural details, while T images are more robust under low illumination and nighttime conditions. However, most multimodal anti-UAV methods still assume well-aligned inputs. In practical dual-sensor systems, asynchronous imaging, sensor displacement, and rapid target motion often cause substantial cross-modal misalignment. Existing weakly alignment-aware methods, such as OAFA [20] and DCNet [21], attempt to mitigate this issue, but they mainly focus on limited perturbations or local alignment. Large spatial displacement and severe modality imbalance remain underexplored in RGB–T UAV detection.

2.3. Learnable cross-modal alignment in multimodal detection

Learnable cross-modal alignment has been widely studied in multimodal 3D detection, as calibration errors, temporal misalignment, and projection uncertainty can degrade cross-modal correspondence [38,39]. GraphAlign and GraphAlign++ [40,41] improve LiDAR–camera correspondence through graph matching, while GraphBEV and GraphBEV++ [42,43] explore learnable alignment in the BEV space. SparseInteraction [44] uses semantic-guided interaction to alleviate positional ambiguity in radar–camera fusion. Related graph-based modeling is also used in LiDAR-only detection, such as VP-Net [45]. Although these studies focus mainly on heterogeneous 3D sensors, their common idea of learning cross-modal correspondence is relevant to RFVTD. Existing 3D methods mainly operate on point–image projections, local graph relations, or BEV representations, whereas RFVTD addresses spatial misalignment between visible and thermal image features using deformation-based alignment.

Table 1
Comparison of representative methods for misaligned visible–thermal detection.

Method	Aligned feature	Geometric modeling	Alignment refinement	Misalignment regime
CMA-Det	Sensed feature	Global translation/scale	No iterative refinement	Position/scale
OAFA	Invariant + specific	Offset-guided sampling	Local deformation correction	Weak
RFVTD	Shared only	Dense deformation	Bidirectional progressive refinement	Large/non-uniform

2.4. Registration-free visible–thermal detection

Registration-free visible–thermal detection relaxes the need for strict geometric correspondence between RGB and T inputs. Traditional registration methods, such as SIFT- or homography-based approaches [15], are often unreliable in UAV scenarios because of sensor noise, parallax, weak cross-modal texture, and real-time constraints. Recent methods instead handle RGB–T misalignment within the detection framework through cross-modal interaction and feature-level alignment [20,22,23,46]. CMA-Det [22] enhances cross-modal correspondence and regresses translation and scale parameters to warp the sensed feature to the reference modality. This global transformation corrects overall position and scale discrepancies but cannot capture spatially varying local displacement. OAFA [20] decomposes multimodal features into modality-invariant and modality-specific representations, estimates a basic offset from the invariant features, and uses it to guide deformable sampling. While this reduces modality discrepancy during offset estimation, the subsequent alignment still relies on the basic offset for local correction. OAFA is primarily designed for weak misalignment, while long-range correspondence and progressive dense deformation refinement under large displacement are not explicitly modeled.

Compared with these methods, RFVTD targets large and spatially non-uniform cross-modal displacement. It separates modality-invariant from modality-specific information and aligns only the shared representation, reducing the risk of interpreting intrinsic modality differences as geometric misalignment. RFVTD further uses non-local cross-modal correspondences to initialize dense deformation and progressively refines residual displacement in both directions. As summarized in Table 1, this formulation offers more flexible correction of large, spatially varying misalignment than global parametric warping or local offset-guided sampling.

3. Method

3.1. Overview

Fig. 1 illustrates the overall architecture of the proposed RFVTD framework. Given an RGB image I_R and a T image I_T , RFVTD first extracts multi-scale modality-specific features $\{F_m^i\}_{m \in \{R,T\}, i=1}^L$ and then processes them with three components: the Shared–Private Decomposition Module (SPDM), the Adaptive Modality Semantic Association and Enhancement Module (AMSEM), and the Dual-Supervised Feature Alignment (DSFA) module.

Throughout this section, $m \in \{R, T\}$ denotes a modality, where R and T refer to RGB and T, respectively. For a modality m , $\bar{m}$ denotes its opposite modality, i.e., $\bar{R} = T$ and $\bar{T} = R$.

SPDM separates each modality feature into shared and private components. The shared component captures modality-invariant structural cues for cross-modal association and alignment, while the private component preserves modality-specific information, such as RGB textures and T thermal responses. AMSEM then builds coarse bidirectional semantic associations between the shared RGB and T features. These associations provide non-local cross-modal cues and target-aware feature enhancement. DSFA further converts the association maps into initial deformation priors and performs progressive residual deformation refinement in both directions, avoiding direct estimation of large displacement from scratch. The aligned cross-modal shared features are finally concatenated with their corresponding private features and fed

into two parallel FPN-based detection heads. Under dual-modality detection supervision, RFVTD jointly learns modality-invariant representation, coarse semantic association, and detection-driven residual alignment for registration-free visible–thermal UAV detection.

3.2. Shared–private decomposition module

In unregistered visible–thermal detection, directly aligning backbone features is suboptimal. RGB and T features contain both modality-invariant structures, such as object geometry and coarse layout, and modality-specific cues, such as RGB textures and thermal radiation patterns. Forcing these heterogeneous cues into a single aligned space may introduce negative transfer. To reduce this interference, we introduce the Shared–Private Decomposition Module (SPDM), which separates each modality feature into a shared component and a private component.

Given the feature F_m^i from modality m at pyramid level i , SPDM uses two lightweight projection heads to obtain

$$F_m^{sh,i} = G_{sh}^i(F_m^i), \quad F_m^{pr,i} = G_{pr}^i(F_m^i). \quad (1)$$

where $F_m^{sh,i}$ and $F_m^{pr,i}$ denote the shared and private features, respectively. The shared feature is used for cross-modal semantic association and residual alignment, while the private feature bypasses the alignment process and is later fused with the aligned shared representation for detection. This design enables the model to align modality-invariant structures while preserving modality-specific discriminative information.

To constrain the decomposition, SPDM is optimized with three objectives. First, a reconstruction branch recovers the original feature from the concatenated shared and private features:

$$\bar{F}_m^i = R^i([F_m^{sh,i} \parallel F_m^{pr,i}]). \quad (2)$$

The reconstruction loss is defined as

$$\mathcal{L}_{rec} = \sum_{i=1}^L \sum_{m \in \{R,T\}} \|\bar{F}_m^i - F_m^i\|_2^2. \quad (3)$$

This term prevents degenerate decomposition and encourages information preservation.

Second, we impose a decorrelation constraint to reduce redundancy between the shared and private spaces. Let $X_m^{sh,i} \in \mathbb{R}^{C_i \times N_i}$ and $X_m^{pr,i} \in \mathbb{R}^{C_i \times N_i}$ denote the flattened shared and private features, where $N_i = H_i W_i$. Before computing the constraint, we normalize each channel by removing its mean and dividing by its standard deviation, obtaining $\tilde{X}_m^{sh,i}$ and $\tilde{X}_m^{pr,i}$. The cross-correlation matrix is computed as

$$C_m^i = \frac{1}{N_i} \tilde{X}_m^{sh,i} (\tilde{X}_m^{pr,i})^\top. \quad (4)$$

The decorrelation loss is defined as

$$\mathcal{L}_\perp = \sum_{i=1}^L \sum_{m \in \{R,T\}} \|C_m^i\|_F^2. \quad (5)$$

This loss suppresses cross-correlation between the shared and private branches and avoids being minimized by simply shrinking feature magnitudes.

Third, we use an adversarial modality classifier to encourage modality invariance in the shared space. The pooled shared feature

$$z_m^i = \text{GAP}(F_m^{sh,i}) \quad (6)$$

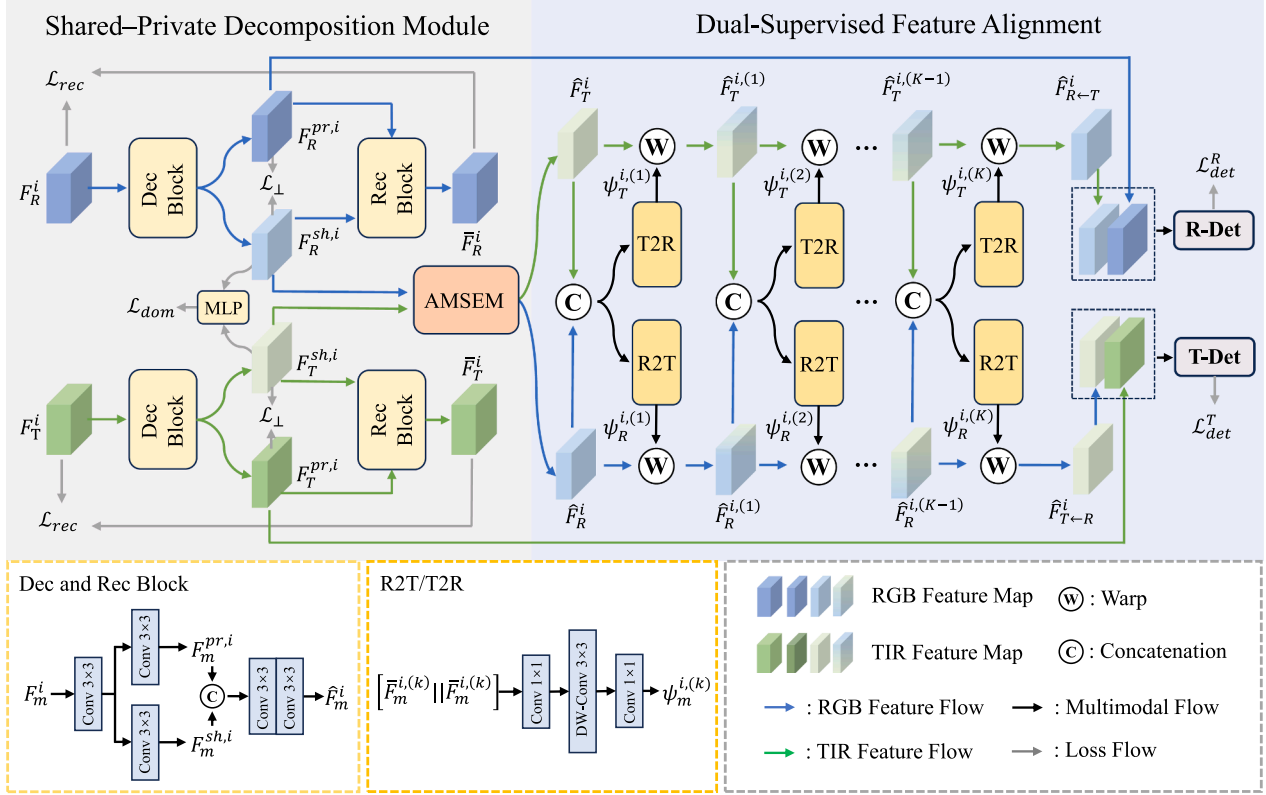


Fig. 1. Overall architecture of the proposed RFVTD framework, illustrated at pyramid level i . SPDM decomposes each modality feature into shared and private components with reconstruction, decorrelation, and adversarial constraints. AMSEM establishes coarse bidirectional semantic associations on a downsampled feature grid, aggregates non-local cross-modal cues, and derives initial deformation priors for subsequent alignment. DSFA refines residual deformation in both RGB-to-T and T-to-RGB directions under dual-modality detection supervision. The aligned cross-modal shared features are finally concatenated with their corresponding private features and fed into two parallel FPN-based detection heads.

is fed to a discriminator D to predict the modality label y_m . The discriminator is trained with the binary cross-entropy loss

$$\mathcal{L}_{dom} = - \sum_{i=1}^L \sum_{m \in \{R,T\}} \left[y_m \log D(z_m^i) + (1 - y_m) \log (1 - D(z_m^i)) \right]. \quad (7)$$

A gradient reversal layer is inserted before D . Thus, the discriminator is optimized by minimizing $\mathcal{L}_{dom}$, while the shared feature extractor receives reversed gradients from the same loss and learns modality-invariant representations.

The SPDM objective is

$$\mathcal{L}_{SPDM} = \lambda_{rec} \mathcal{L}_{rec} + \lambda_{\perp} \mathcal{L}_{\perp} + \lambda_{adv} \mathcal{L}_{dom}. \quad (8)$$

where λ_{rec} , $\lambda_{\perp}$, and λ_{adv} are balancing weights for the reconstruction, decorrelation, and adversarial modality objectives, respectively.

Through these constraints, SPDM encourages the shared branch to retain modality-invariant structural cues and the private branch to preserve modality-specific details. The resulting shared features provide a reliable basis for AMSEM to establish cross-modal semantic associations, while the private features complement the final detection representation.

3.3. Adaptive modality semantic association and enhancement module

After SPDM, the shared features contain modality-invariant cues that are more suitable for cross-modal interaction. However, under large spatial displacement, local convolutional predictors are better suited to residual correction than to establishing long-range correspondences from scratch. Moreover, UAV targets are usually small and easily suppressed by clutter, while the reliability of RGB and T observations varies

with illumination. To address these issues, we propose the Adaptive Modality Semantic Association and Enhancement Module (AMSEM), which constructs coarse bidirectional semantic associations and enhances target-aware features before DSFA performs association-based initialization and residual alignment.

Given the shared features $\{F_m^{sh,i}\}_{m \in \{R,T\}}$ at pyramid level i , where

$$F_m^{sh,i} \in \mathbb{R}^{C_i \times H_i \times W_i}, \quad (9)$$

AMSEM outputs refined shared features $\{\hat{F}_m^i\}_{m \in \{R,T\}}$ and bidirectional association maps $\{A_{m \rightarrow \bar{m}}^i\}_{m \in \{R,T\}}$.

3.3.1. Cross-modal semantic association

To reduce the cost of global association, we compute cross-modal association on a downsampled feature grid and then upsample the associated features to the original pyramid resolution. Specifically, the shared features are first downsampled as

$$\tilde{F}_m^{sh,i} = \mathcal{P}_i(F_m^{sh,i}), \quad (10)$$

where $\mathcal{P}_i(\cdot)$ denotes spatial pooling or strided projection. In our implementation, $\mathcal{P}_i(\cdot)$ is implemented by adaptive average pooling to a fixed spatial size, followed by a 1×1 projection. The downsampled features satisfy

$$\tilde{F}_m^{sh,i} \in \mathbb{R}^{C_i \times h_i \times w_i}, \quad \tilde{N}_i = h_i w_i, \quad (11)$$

where $\tilde{N}_i \ll H_i W_i$.

We then project the downsampled shared features into a compact embedding space:

$$\begin{aligned} \tilde{Q}_m^i &= W_Q^m(\tilde{F}_m^{sh,i}), \\ \tilde{K}_m^i &= W_K^m(\tilde{F}_m^{sh,i}), \\ \tilde{V}_m^i &= W_V^m(\tilde{F}_m^{sh,i}). \end{aligned} \quad (12)$$

where W_Q^m , W_K^m , and W_V^m are 1×1 convolutional projections. After flattening the spatial dimensions, $\tilde{Q}_m^i, \tilde{K}_m^i, \tilde{V}_m^i \in \mathbb{R}^{\tilde{N}_i \times d_i}$.

For each modality m , the cross-modal association map from m to $\bar{m}$ is computed by global attention on the downsampled grid:

$$A_{m \rightarrow \bar{m}}^i = \text{Softmax} \left(\frac{\tilde{Q}_m^i (\tilde{K}_{\bar{m}}^i)^\top}{\sqrt{d_i}} \right). \quad (13)$$

The softmax is applied along the target-modality spatial dimension. $A_{m \rightarrow \bar{m}}^i(x, y)$ measures the semantic relevance between position x in modality m and position y in the opposite modality $\bar{m}$ on the downsampled grid.

The associated cross-modal feature from $\bar{m}$ to m is first obtained on the downsampled grid:

$$\tilde{F}_{\bar{m} \rightarrow m}^i = W_O^m (\text{Reshape} (A_{m \rightarrow \bar{m}}^i \tilde{V}_{\bar{m}}^i)), \quad (14)$$

where W_O^m restores the channel dimension to C_i . The associated features are then upsampled to the original pyramid resolution:

$$\tilde{F}_{\bar{m} \rightarrow m}^i = \text{Up} (\tilde{F}_{\bar{m} \rightarrow m}^i). \quad (15)$$

These features provide non-local semantic correspondence cues from the opposite modality while keeping the association cost tractable.

3.3.2. Semantic enhancement and reliability-aware weighting

To strengthen small UAV responses, we inject cross-modal association cues into the shared features:

$$F_m^{a,i} = F_m^{sh,i} + \lambda_a \tilde{F}_{\bar{m} \rightarrow m}^i, \quad (16)$$

where λ_a controls the association strength.

We then propagate high-level semantic cues from level $i + 1$ to level i . For $i < L$, the shared features from the next pyramid level are concatenated as

$$F_{cat}^{i+1} = [F_R^{sh,i+1} \parallel F_T^{sh,i+1}], \quad (17)$$

and encoded by multi-head self-attention:

$$F_G^{i+1} = \text{MHSA} (F_{cat}^{i+1}) + F_{cat}^{i+1}. \quad (18)$$

After splitting F_G^{i+1} into modality-specific branches, semantic guidance maps are generated by

$$S_m^i = \sigma \left(\text{Conv}_{3 \times 3} \left(\text{Up} (F_{G,m}^{i+1}) \right) \right). \quad (19)$$

For the top pyramid level, we omit top-down guidance and set $S_m^L = 0$. The association-aware features are enhanced as

$$\tilde{F}_m^i = (1 + \beta S_m^i) \odot F_m^{a,i}, \quad (20)$$

where β controls the enhancement strength.

Finally, we estimate modality reliability using both image-level statistics and association confidence. The scene descriptor is

$$q = [\mu(\text{gray}(I_R)), \text{std}(\text{gray}(I_R)), \mu(I_T), \text{std}(I_T)]. \quad (21)$$

For each modality m , the association confidence is computed from the entropy of the downsampled association distribution:

$$\tilde{C}_{m \rightarrow \bar{m}}^i(x) = 1 - \frac{H(A_{m \rightarrow \bar{m}}^i(x, \cdot))}{\log \tilde{N}_i}. \quad (22)$$

The modality weights are given by

$$u^i = [q \parallel \text{GAP}(\tilde{C}_{R \rightarrow T}^i) \parallel \text{GAP}(\tilde{C}_{T \rightarrow R}^i)], \quad (23)$$

$$[g_R^i, g_T^i] = \text{Softmax} (\text{MLP}(u^i)). \quad (24)$$

Here, g_R^i and g_T^i are level-wise scalar weights and are broadcast to all spatial positions. The final AMSEM outputs are

$$\hat{F}_m^i = g_m^i \tilde{F}_m^i + g_{\bar{m}}^i \tilde{F}_{\bar{m} \rightarrow m}^i. \quad (25)$$

AMSEM therefore returns

$$\{\{\hat{F}_m^i\}_{m \in \{R,T\}}, \{A_{m \rightarrow \bar{m}}^i\}_{m \in \{R,T\}}\}_{i=1}^L, \quad (26)$$

which are passed to DSFA for association-based initialization and progressive residual alignment.

3.4. Dual-supervised feature alignment

RGB and T features are often spatially misaligned due to sensor displacement, viewpoint difference, and temporal offset. After AMSEM, coarse cross-modal correspondences have been established through bidirectional semantic association. However, these association maps provide only coarse non-local correspondence cues and are not sufficient for accurate localization. We therefore introduce the Dual-Supervised Feature Alignment (DSFA) module to convert the associations into initial deformation priors and then refine the remaining deformation in a progressive and detection-driven manner.

Given the AMSEM outputs at pyramid level i , including the refined shared features $\{\hat{F}_m^i\}_{m \in \{R,T\}}$ and the bidirectional association maps $\{A_{m \rightarrow \bar{m}}^i\}_{m \in \{R,T\}}$, DSFA first performs association-based initialization. Let $\bar{p}_x \in \mathbb{R}^2$ denote the coordinate of position x on the downsampled association grid. For each modality m , the initial deformation prior from m to $\bar{m}$ is computed as

$$\tilde{\psi}_{m \rightarrow \bar{m}}^{i,0}(x) = \sum_{y=1}^{\tilde{N}_i} A_{m \rightarrow \bar{m}}^i(x, y) (\bar{p}_y - \bar{p}_x). \quad (27)$$

The prior is then upsampled to the original pyramid resolution:

$$\psi_{m \rightarrow \bar{m}}^{i,0} = \mathbf{r}_i \odot \text{Up} (\tilde{\psi}_{m \rightarrow \bar{m}}^{i,0}). \quad (28)$$

Here, $\mathbf{r}_i = (W_i/w_i, H_i/h_i)$ is the spatial scaling factor between the downsampled association grid and the original pyramid level, and $\odot$ denotes element-wise multiplication.

The deformation fields are initialized as

$$\psi_{m \rightarrow \bar{m}}^{i,0} = \psi_{m \rightarrow \bar{m}}^{i,0}. \quad (29)$$

The corresponding initially warped features are obtained by

$$\hat{F}_{\bar{m} \leftarrow m}^{i,(0)} = W(\hat{F}_m^i, \psi_{m \rightarrow \bar{m}}^{i,(0)}), \quad (30)$$

where $W(\cdot, \psi)$ denotes differentiable feature warping. The deformation field $\psi_{m \rightarrow \bar{m}}$ is defined as a forward displacement from the grid of modality m to the grid of modality $\bar{m}$. In implementation, $W(\cdot, \psi)$ converts the forward displacement into a backward sampling grid for differentiable bilinear warping. Thus, $\hat{F}_{\bar{m} \leftarrow m}^{i,(0)}$ denotes the shared feature of modality m warped toward the feature space of modality $\bar{m}$.

At refinement stage $k \in \{1, \dots, K\}$, DSFA estimates residual deformation fields rather than full displacements. For each modality m , the residual is predicted by a lightweight convolutional head:

$$\Delta \psi_{m \rightarrow \bar{m}}^{i,(k)} = P_{m \rightarrow \bar{m}}^{i,(k)} ([\hat{F}_{\bar{m} \leftarrow m}^{i,(k-1)} \parallel \hat{F}_{\bar{m}}^i \parallel \psi_{m \rightarrow \bar{m}}^{i,(k-1)}]). \quad (31)$$

The deformation fields are updated in a residual manner:

$$\psi_{m \rightarrow \bar{m}}^{i,(k)} = \psi_{m \rightarrow \bar{m}}^{i,(k-1)} + \Delta \psi_{m \rightarrow \bar{m}}^{i,(k)}. \quad (32)$$

The aligned features are then updated by warping the original AMSEM-enhanced shared features with the refined deformation fields:

$$\hat{F}_{\bar{m} \leftarrow m}^{i,(k)} = W(\hat{F}_m^i, \psi_{m \rightarrow \bar{m}}^{i,(k)}). \quad (33)$$

After K refinement stages, the final aligned shared features are

$$\hat{F}_{\bar{m} \leftarrow m}^i = \hat{F}_{\bar{m} \leftarrow m}^{i,(K)}. \quad (34)$$

This formulation separates coarse correspondence estimation from residual geometric refinement. AMSEM provides non-local semantic association maps, while DSFA converts them into deformation priors and focuses on local residual correction. Thus, the lightweight predictors are not required to recover large cross-modal displacement from scratch.

To preserve modality-specific discrimination while avoiding redundant shared representations, we fuse the aligned cross-modal shared feature with the corresponding private feature from SPD. The aligned shared feature provides modality-invariant structural cues in the target feature space, whereas the private feature preserves target-modality-specific details:

$$Z_m^i = [\hat{F}_{m \leftarrow \bar{m}}^i \parallel F_m^{pr,i}]. \quad (35)$$

The fused multi-scale features $\{Z_m^i\}_{m \in \{R, T\}}$ are fed into two parallel FPN-based detection heads. The RGB branch is supervised by RGB annotations, and the T branch is supervised by T annotations. The dual-modality detection loss is defined as

$$\mathcal{L}_{det} = \sum_{m \in \{R, T\}} \mathcal{L}_{det}^m(O_m, Y_m), \quad (36)$$

where each branch contains a classification loss and a regression loss:

$$\mathcal{L}_{det}^m = \mathcal{L}_{cls}^m + \lambda_{reg} \mathcal{L}_{reg}^m. \quad (37)$$

To stabilize both the association-based initial priors and the DSFA-refined deformation fields, we impose an alignment consistency loss over all stages, including $k = 0$:

$$\mathcal{L}_{align} = \sum_{i=1}^L \sum_{k=0}^K \sum_{m \in \{R, T\}} \left\| \hat{F}_{\bar{m}-m}^{i,(k)} - \hat{F}_{\bar{m}}^i \right\|_2^2. \quad (38)$$

The final DSFA objective is jointly driven by dual-modality detection supervision and alignment consistency. The detection loss preserves target discriminability, while the alignment loss regularizes the deformation process. As a result, DSFA learns task-oriented cross-modal feature alignment without requiring pixel-level registration supervision.

3.5. Overall training objective

The whole framework is trained end-to-end with

$$\mathcal{L} = \mathcal{L}_{det} + \lambda_{align} \mathcal{L}_{align} + \lambda_{SPDM} \mathcal{L}_{SPDM}. \quad (39)$$

Here, λ_{align} and λ_{SPDM} balance the alignment constraint and the shared-private decomposition objective, respectively. AMSEM is optimized through the detection and alignment objectives, since its association maps and deformation priors affect both the warped features and the final detection outputs.

4. Experiments

4.1. Datasets

To comprehensively evaluate RFVTD, we conduct experiments on four publicly available RGB-T benchmarks, covering both UAV-specific and general object detection tasks with varying registration conditions.

4.1.1. Anti-UAV detection

We construct a detection-oriented split from the Anti-UAV dataset [8], a large-scale multi-modal UAV tracking benchmark containing 318 pairs of unaligned RGB-T video sequences captured at 25 FPS. We follow the original train, validation, and test partitions provided by Anti-UAV. To reduce temporal redundancy, we sample one frame every 25 frames from each sequence. After preprocessing, the resulting split contains 5,710 RGB-T image pairs for training, 2,340 pairs for validation, and 3,202 pairs for testing. All compared methods are trained and evaluated using the same processed split and identical preprocessing protocol. Therefore, the reported results on Anti-UAV should be interpreted as detection performance on our constructed Anti-UAV detection split, rather than as results under the original tracking benchmark protocol.

4.1.2. General RGB-T detection datasets

To further examine the generalization capability of RFVTD beyond UAV-specific tasks, we evaluate it on three representative RGB-T detection datasets, which span a spectrum of modality alignment levels:

DVTD [22]: A recently released UAV-mounted benchmark focused on misaligned visible-thermal object detection. It consists of 2,179 image pairs across 54 scenes with 16 challenging attributes, all exhibiting significant misalignment. Following the official split, 1,606 pairs are used for training and 573 for testing. This dataset is critical for testing robustness under complex **unregistered conditions**.

DroneVehicle [23]: A large-scale dataset for vehicle detection captured by UAVs in diverse real-world environments. 28,439 RGB-T pairs with 953,087 oriented bounding boxes for five vehicle categories. Although pre-registration is applied, about 35% of samples still show residual misalignment [20]. It enables evaluation of RFVTD under **weakly misaligned conditions**.

LLVIP [24]: A standard aligned RGB-T benchmark collected in low-light conditions, consisting of 15,488 manually aligned image pairs (12,025 for training, 3,463 for testing). It is used to verify whether RFVTD maintains strong performance even under **aligned conditions**.

Together, these datasets form a comprehensive evaluation suite, enabling assessment of RFVTD's robustness from severely misaligned UAV imagery to well-aligned low-light scenarios.

4.2. Experimental settings

We implement RFVTD in PyTorch and conduct all experiments on a single NVIDIA A100 GPU with 40 GB of memory. The model is trained for 400 epochs using SGD with a momentum of 0.9 and a weight decay of 5×10^{-4} . The initial learning rate is set to 0.01 and scheduled with cosine annealing, while the batch size is fixed at 16. All input images are resized to 640×640 . In DSFA, we employ dual-path deformation alignment with $K = 3$ residual refinement stages. For methods with available implementations, we reproduce them under the same preprocessing and evaluation protocol. When official implementations are unavailable or full reproduction is infeasible, we report the results published in the original papers under comparable key settings.

The key hyperparameters of RFVTD are set as follows. For AMSEM, the downsampled association grid is set to $h_i \times w_i = 16 \times 16$. The association injection coefficient and semantic enhancement coefficient are set to $\lambda_a = 1.0$ and $\beta = 1.0$, respectively. For DSFA, the number of residual refinement stages is set to $K = 3$. For the training objective, we set $\lambda_{align} = 0.1$, $\lambda_{SPDM} = 1.0$, $\lambda_{rec} = 1.0$, $\lambda_{\perp} = 0.1$, and $\lambda_{dom} = 0.01$.

We evaluate the performance of RFVTD and all comparison methods using standard object detection metrics. The primary evaluation criterion is the mean Average Precision (mAP), calculated as the mean of Average Precision (AP) across all object categories. Unless otherwise specified, mAP denotes AP averaged over categories at an IoU threshold of 0.5.

During training, RFVTD adopts dual-branch supervision where both RGB and thermal heads are optimized with their respective annotations. At inference, we activate a single modality-specific detector depending on the evaluation setting and report the corresponding mAP. Unless otherwise specified, RFVTD reports the T-branch detection results for RGB-T evaluation.

4.3. Comparison methods

To comprehensively assess the performance of the proposed RFVTD framework, we compare it with a range of representative unimodal and multimodal object detection methods across two tasks: UAV detection and general RGB-T object detection.

4.3.1. Anti-UAV detection

For the Anti-UAV dataset, we evaluate both unimodal and multimodal detection approaches.

Unimodal detectors include: YOLOv5s [47], Faster R-CNN [48], RetinaNet [49], and Oriented R-CNN [50]. These methods are trained and evaluated separately on RGB and T inputs to examine the modality-specific detection capability of each sensor.

Multimodal methods include early-fusion, cross-modal fusion, and misalignment-aware detection baselines. YOLOv5s + Add and YOLOv5s + CMX [51] are adopted as early-fusion baselines, where RGB and thermal features are combined through simple addition or channel mixing. CFT [33] is included as a Transformer-based cross-modal fusion

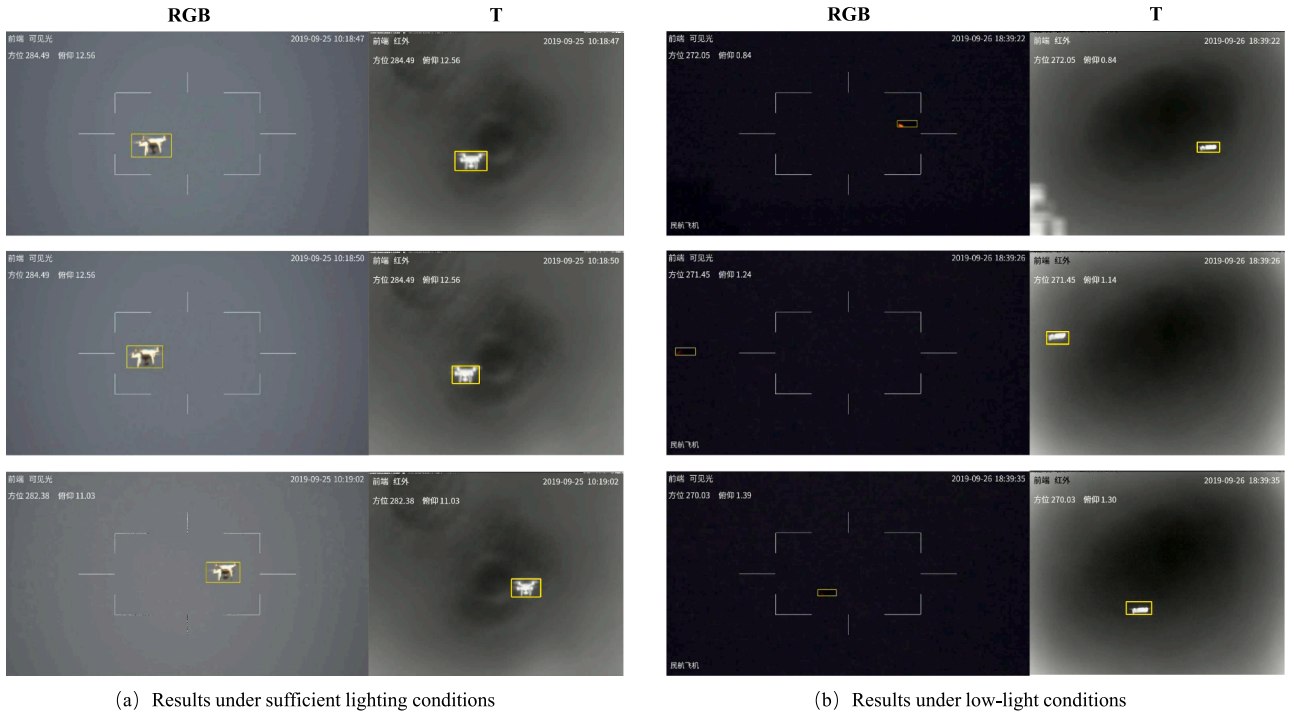


Fig. 2. Qualitative UAV detection results under different illumination conditions. (a) Well-lit scenes where UAV targets are clearly distinguishable. (b) Low-light scenes where RGB features degrade severely, yet RFVTD accurately localizes UAVs.

method. UA-CMDet [23] and CMA-Det [22] are selected as representative alignment-aware RGB-T detection methods, since they explicitly model modality discrepancy or feature alignment under weakly misaligned inputs. In addition, we include RSDet [52], a recent coarse-to-fine RGB-T fusion detector that removes redundant spectral information and dynamically selects discriminative cross-modal features.

4.3.2. General RGB-T detection

We adopt both unimodal and multimodal object detectors for comparison.

Unimodal baselines include: YOLOv5s, RetinaNet, Faster R-CNN, RoITransformer [53], S2ANet [54], and Oriented R-CNN, all evaluated separately on RGB and thermal inputs.

Multimodal fusion methods include: Halfway Fusion [25], CIAN [28], MBNet [30], and AR-CNN [55], which adopt cross-modality attention or feature blending strategies. In addition, TSFADet [46], C²Former [31], and OAFA [20] are included as stronger RGB-T fusion baselines that address weak spatial misalignment or modality calibration through adaptive fusion and alignment-aware modules.

4.4. Evaluation results on Anti-UAV detection

4.4.1. Quantitative results

Table 2 presents the quantitative comparison on the Anti-UAV detection split. Several observations can be made.

First, unimodal detectors show limited robustness when relying on a single sensor. Among RGB-based methods, YOLOv5s achieves the best performance with 51.2% mAP, while the best thermal-only result is 48.6% mAP. This indicates that neither RGB nor thermal imagery alone can consistently handle the challenging UAV scenarios, where targets are usually small and easily affected by illumination degradation, thermal blur, and complex background interference.

Second, simple early-fusion baselines do not necessarily improve detection performance. YOLOv5s + Add obtains only 45.1% mAP, which is even lower than the RGB-only YOLOv5s baseline. YOLOv5s + CMX improves the result to 47.6%, but it still remains inferior to the strongest

Table 2

Performance comparison of different methods on the Anti-UAV dataset.

Methods	Modality	mAP(%) ↑
Faster R-CNN [48]	RGB	46.4
RetinaNet [49]		39.3
YOLOv5s [47]		51.2
Oriented R-CNN [50]		49.8
Faster R-CNN [48]	T	45.5
RetinaNet [49]		40.4
YOLOv5s [47]		48.6
Oriented R-CNN [50]		46.4
YOLOv5s + Add	RGB + T	45.1
YOLOv5s + CMX [51]		47.6
CFT [33]		49.7
UA-CMDet [23]		52.5
CMA-Det [22]		53.9
RSDet [52]		53.7
RFVTD (Ours)		56.1

unimodal detector. These results suggest that directly combining RGB and thermal features without explicitly considering cross-modal discrepancy may introduce feature conflicts, especially when the two modalities are spatially misaligned.

Third, methods that explicitly enhance cross-modal interaction or alleviate modality discrepancy achieve better performance. CFT, UA-CMDet, and CMA-Det improve the mAP to 49.7%, 52.5%, and 53.9%, respectively, demonstrating the importance of modeling cross-modal relationships under misaligned RGB-T inputs. RSDet, a recent coarse-to-fine RGB-T fusion detector, achieves 53.7% mAP. Although RSDet introduces redundant spectrum removal and dynamic feature selection to suppress interfering modality information, its performance is slightly lower than CMA-Det on this UAV detection split. This indicates that feature purification and selection alone are not sufficient to fully address the severe spatial displacement and small-target localization challenges in unregistered UAV scenes.

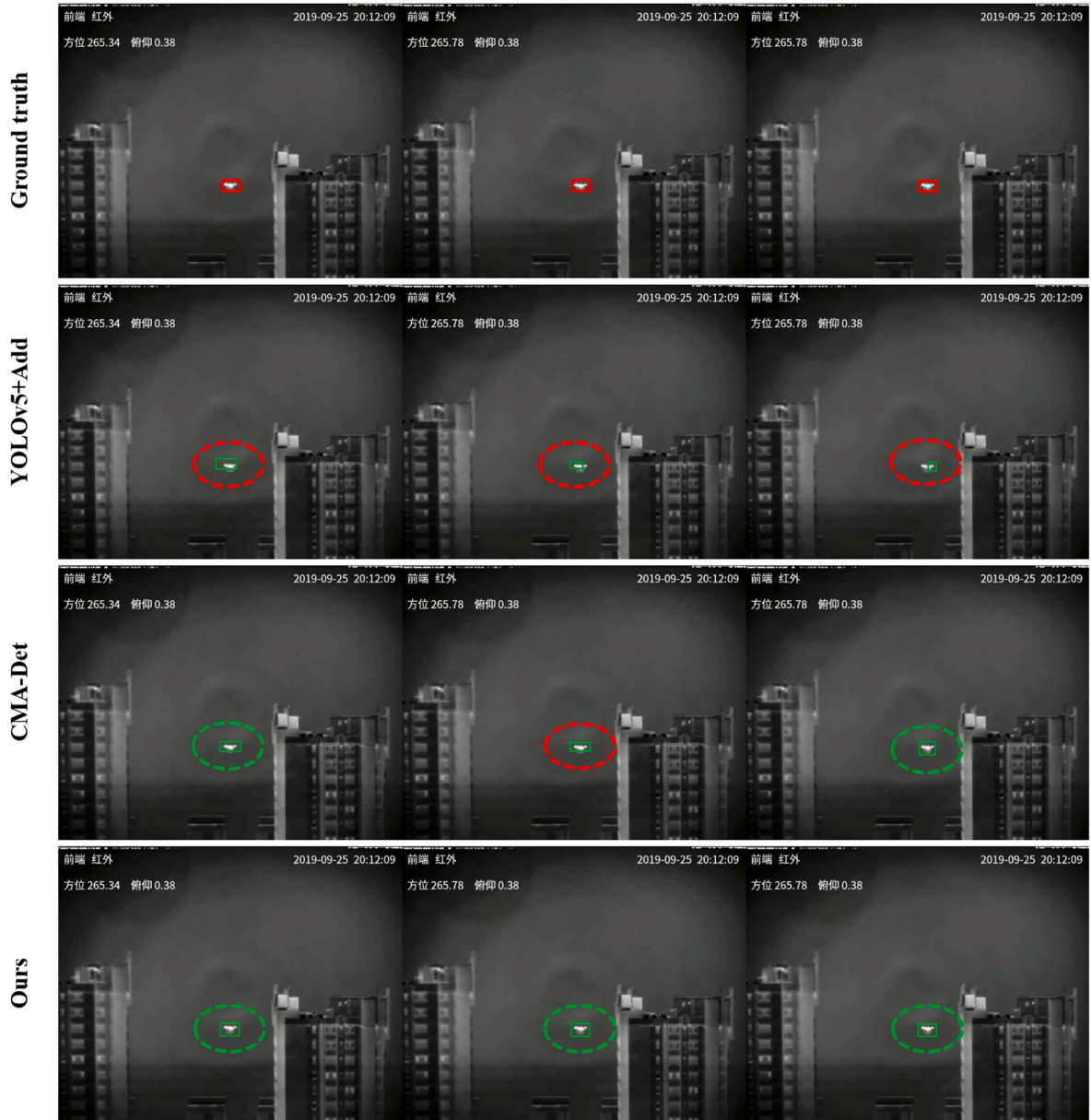


Fig. 3. Qualitative comparison of UAV detection under low-light conditions with misaligned inputs. Results of YOLOv5s + Add, CMA-Det, and the proposed RFVTD are shown. Correct detections are highlighted with green dashed circles, while false detections are marked in red dashed circles.

Finally, the proposed RFVTD achieves the best performance with 56.1% mAP. It surpasses CMA-Det by 2.2% and RSDet by 2.4%. The improvement is mainly attributed to the proposed registration-free coarse-to-fine feature interaction design. Specifically, SPDM separates modality-invariant shared cues from modality-specific private cues, AM-SEM establishes non-local semantic associations and enhances UAV-relevant responses, and DSFA progressively refines residual cross-modal deformation under dual-modality detection supervision. These results demonstrate that RFVTD can more effectively exploit RGB-T complementarity while reducing negative transfer caused by severe cross-modal misalignment.

4.4.2. Qualitative results

The quantitative results show that RFVTD achieves clear advantages over both naive fusion and alignment-aware baselines on the Anti-UAV detection split. To further examine where these improvements come from, we provide qualitative comparisons under representative UAV detection scenarios, including normal illumination, low-light

conditions, and severe cross-modal misalignment. Figs. 2–4 compare YOLOv5s + Add, CMA-Det, and RFVTD.

Fig. 2 shows detection examples under well-lit and low-light conditions. In well-lit scenes, all methods can capture some UAV cues, but RFVTD produces more stable localization. Under low-light conditions, RGB features become weak and noisy, making naive fusion prone to missed detections or unstable responses. RFVTD still localizes UAV targets accurately by exploiting complementary thermal cues and suppressing unreliable modality-specific responses. This indicates that the proposed shared-private decomposition and semantic association help preserve useful cross-modal information under illumination degradation.

Figs. 3 and 4 further present examples under low-light conditions with misaligned inputs. YOLOv5s + Add suffers from direct fusion of unaligned features and often produces false detections. CMA-Det improves over the baseline but may still miss small UAV structures when T blur, weak visible cues, or local misalignment occurs. In contrast, RFVTD yields more complete and accurate detections across both modalities. This improvement is mainly attributed to three designs: SPDM sepa-

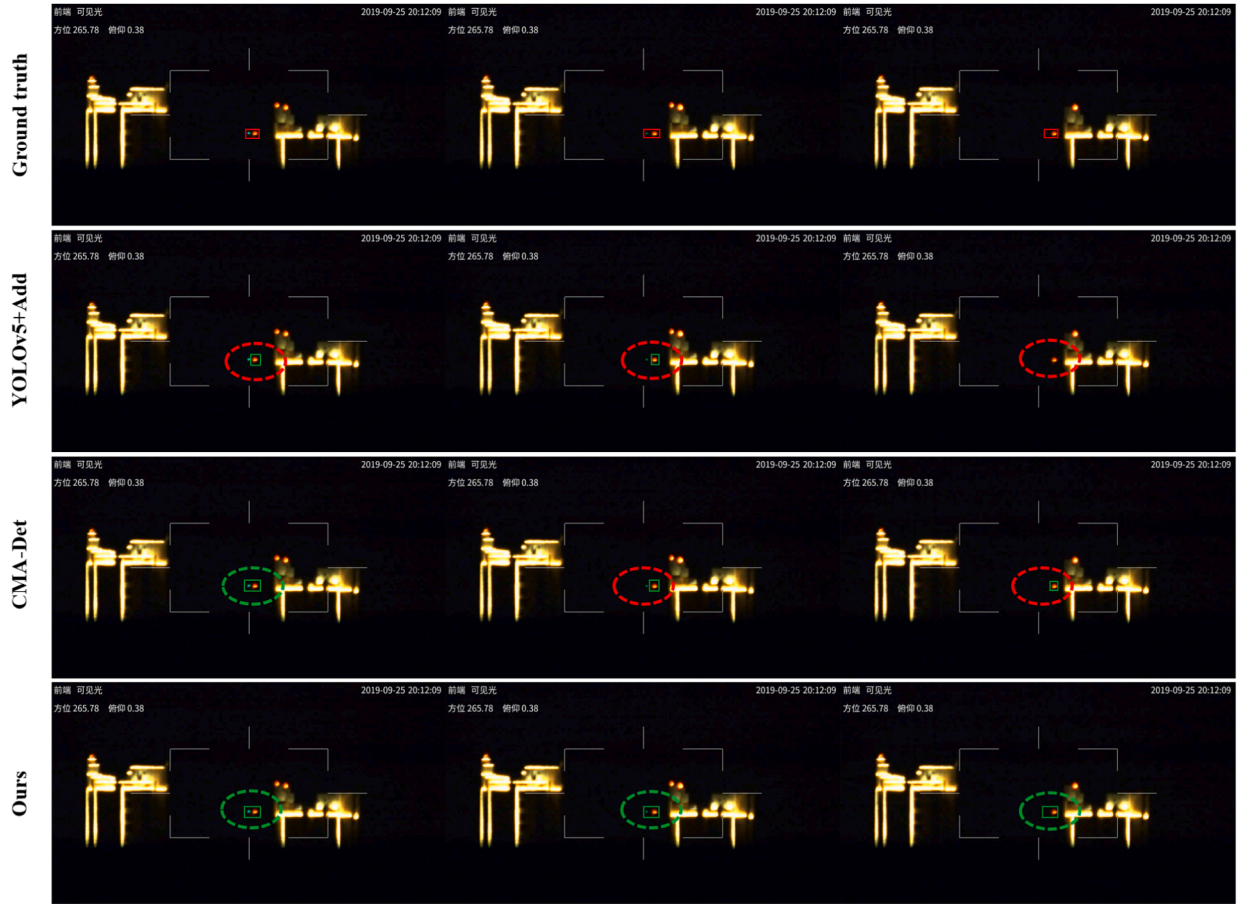


Fig. 4. Qualitative comparison of UAV detection under low-light visible conditions with misaligned inputs. Results of YOLOv5s + Add, CMA-Det, and the proposed RFVTD are shown. Correct detections are highlighted with green dashed circles, while false detections are marked in red dashed circles.

Table 3

Performance comparison on the DVTOD dataset under severe cross-modal misalignment.

Methods	Modality	Person	Car	Bicycle	mAP(%) $\uparrow$
YOLOv5s [47]	RGB	23.3	50.2	26.7	33.4
YOLOv8 [56]		28.4	52.7	24.7	35.3
YOLOv5s [47]	T	86.2	75.9	86.2	82.8
YOLOv8 [56]		86.2	75.6	88.1	83.3
YOLOv5s + Add	RGB + T	88.8	74.3	74.6	79.2
YOLOv5s + CMX [51]		88.9	75.9	79.6	81.6
CMA-Det [22]		90.3	81.6	83.1	85.0
RFVTD (Ours)		91.2	84.4	85.7	87.1

rates modality-invariant and modality-specific representations, AMSEM establishes coarse semantic associations and enhances target-aware responses, and DSFA progressively refines residual cross-modal deformation.

The qualitative comparisons further support the quantitative results. RFVTD shows stronger robustness under low illumination, T blur, and cross-modal misalignment, demonstrating that its registration-free design can better exploit RGB-T complementarity in challenging UAV detection scenarios.

4.5. Evaluation results on general datasets

We further evaluate RFVTD on three RGB-T benchmarks with different alignment conditions. DroneVehicle, DVTOD, and LLVIP correspond to weakly aligned, severely misaligned, and well-aligned RGB-T settings, respectively. These datasets cover a wide range of registra-

tion conditions, enabling a more comprehensive analysis of how cross-modal misalignment affects RGB-T fusion. The results are reported in Tables 3–5.

4.5.1. DroneVehicle

DroneVehicle represents a weakly aligned UAV-based RGB-T detection scenario. Thermal-only detectors generally outperform RGB-only detectors, showing the stronger robustness of thermal imagery in low-light and texture-deficient aerial scenes. Multimodal methods further improve performance, with OAFA reaching 79.4% mAP.

RFVTD achieves 80.5% mAP, outperforming OAFA by 1.1%. The gain is moderate but consistent, especially for challenging categories such as freight cars and vans. This suggests that the proposed feature-level association and residual alignment can compensate for residual modality misalignment while preserving discriminative modality-specific cues.

4.5.2. DVTOD

DVTOD contains severely misaligned RGB-T pairs and is therefore a challenging benchmark for registration-free fusion. RGB-only YOLOv5s obtains only 33.4% mAP, whereas T-only detectors reach 82.8–83.3%, indicating that RGB observations are strongly affected by large cross-modal displacement. Fusion methods outperform unimodal baselines, but early-fusion approaches remain limited because they directly combine unaligned features. CMA-Det reaches 85.0% mAP, while RFVTD achieves the best result of 87.1%, surpassing CMA-Det by 2.1%. This verifies the effectiveness of RFVTD under severe misalignment, where AMSEM establishes non-local semantic association and DSFA progressively refines residual cross-modal deformation.

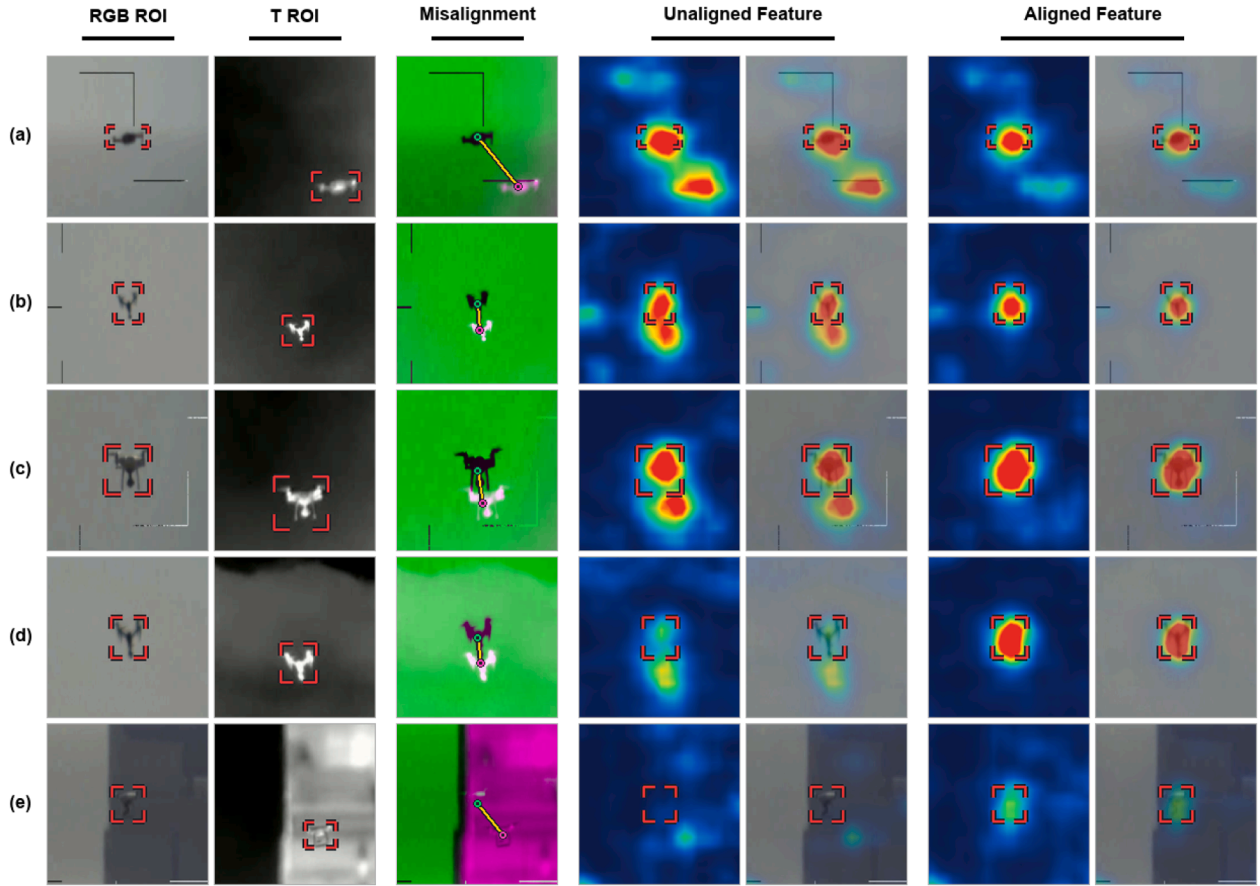


Fig. 5. Visualization of T-to-R feature alignment in DSFA. The RGB and thermal ROIs show clear spatial offsets. For the unaligned and aligned features, the left image shows the feature response map and the right image shows its overlay on the RGB reference image. After alignment, the thermal shared-feature responses become better localized around the corresponding RGB targets. The boxes indicate the ground-truth annotations.

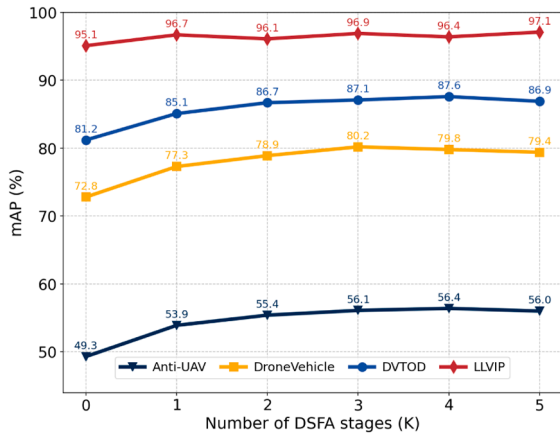


Fig. 6. Impact of DSFA refinement stages K (Q4) across four datasets.

4.5.3. LLVIP

LLVIP is a well-aligned low-light RGB-T benchmark. Since the visible and thermal images are already manually aligned, the advantage of registration-free alignment is less pronounced than on DroneVehicle and DVTOD. In this setting, most fusion methods achieve strong performance. CMA-Det obtains the best result with 97.1% mAP, while RFVTD achieves 96.9% mAP, only 0.2% lower than CMA-Det. Compared with other fusion baselines, RFVTD still remains competitive. It outperforms UA-CMDet by 0.6%, MBNet by 3.7%, and the recent RSDet by 1.1%. The comparison suggests that RFVTD does not sacrifice well-aligned de-

tection performance while being mainly designed for misaligned RGB-T inputs.

Overall, RFVTD achieves the most significant gains on datasets where cross-modal misalignment is evident, especially DVTOD and Anti-UAV, and also performs strongly on weakly aligned DroneVehicle. On the well-aligned LLVIP dataset, RFVTD remains comparable to leading RGB-T fusion methods, although it is slightly lower than CMA-Det. These results show that RFVTD improves robustness to cross-modal misalignment while maintaining competitive RGB-T detection capability under aligned conditions.

4.6. Ablation study

To comprehensively evaluate the effectiveness of our design, we perform ablation studies from four perspectives:

- **Q1: Module contributions.** We isolate the independent and joint effects of SPDM, AMSEM, and DSFA.
- **Q2: SPDM constraints.** We examine the effects of the reconstruction, decorrelation, and adversarial constraints in SPDM.
- **Q3: Alignment direction.** We compare uni- and bidirectional alignment in DSFA to assess their impact on detection accuracy.
- **Q4: Number of refinement stages K .** We vary K in DSFA and evaluate its effect across Anti-UAV, DroneVehicle (weak alignment), DVTOD (severe misalignment), and LLVIP (well-aligned).

4.6.1. Q1. Module contributions

We take YOLOv5s + Add, a simple early-fusion baseline, as the starting point (A0) and progressively incorporate the proposed components on the Anti-UAV dataset. Specifically, we add the Shared-Private

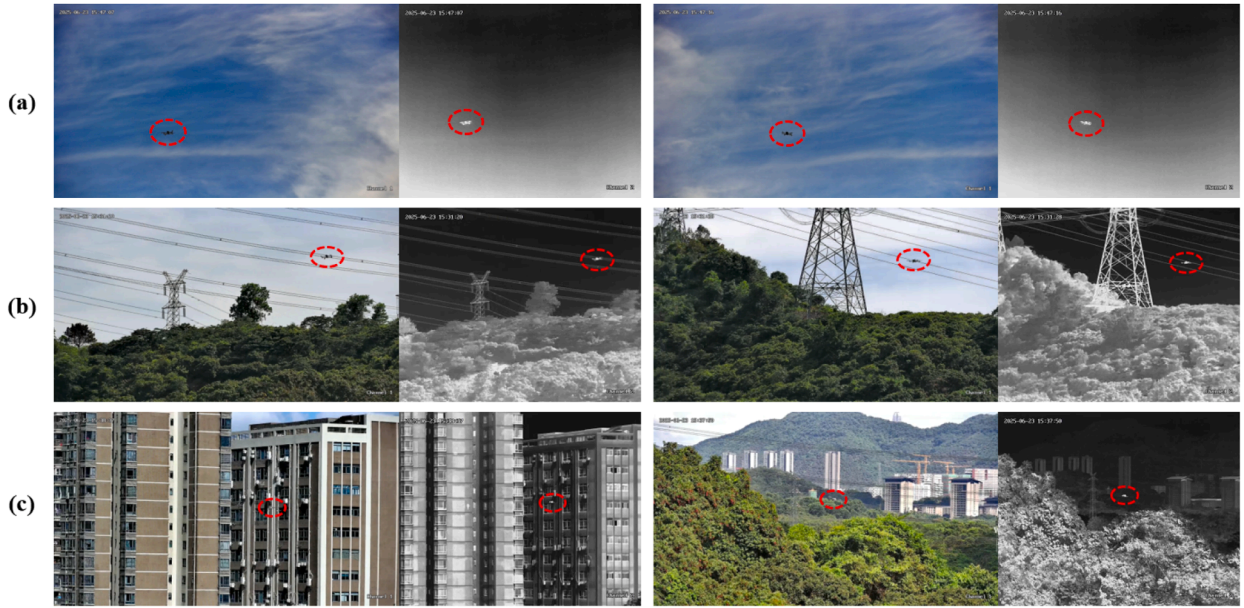


Fig. 7. Example RGB-T image pairs from the collected real-world dataset under three background interference conditions. (a) No background interference, where the UAV is clearly distinguishable from the sky. (b) Weak background interference, where cluttered elements such as power lines introduce mild confusion. (c) Strong background interference, where the UAV is embedded in complex urban or forest backgrounds, significantly increasing the detection difficulty. The UAVs are indicated by red dashed circles.

Table 4

Detection results on the DroneVehicle dataset under weakly aligned conditions. FPS is measured on a single NVIDIA A100 GPU at 640×640 input resolution. All methods use the same OBB evaluation protocol, with mAP computed at IoU@0.5.

Detectors	Modality	Car	Truck	Freight-car	Bus	Van	mAP (%) $\uparrow$	FPS $\uparrow$
RetinaNet [49]	RGB	78.5	34.4	24.1	69.8	28.8	47.1	–
Faster R-CNN [48]		79.0	49.0	37.2	77.0	37.0	55.9	–
Oriented R-CNN [50]		80.1	53.8	41.6	85.4	43.3	60.8	–
S ² A-Net [54]		80.0	54.2	42.2	84.9	43.8	61.0	–
RoITransformer [53]		61.6	55.1	42.3	85.5	44.8	61.6	–
YOLOv5s [47]		78.6	55.3	43.8	87.1	46.0	62.1	–
RetinaNet [49]	T	88.8	35.4	39.5	76.5	32.1	54.5	–
Faster R-CNN [48]		89.4	53.5	48.3	87.0	42.6	64.2	–
Oriented R-CNN [50]		89.8	57.4	53.1	89.3	45.4	67.0	–
S ² A-Net [54]		89.9	54.5	55.8	88.9	48.4	67.5	–
RoITransformer [53]		90.1	60.4	58.9	89.7	52.2	70.3	–
YOLOv5s [47]		90.0	59.5	60.8	89.5	53.8	70.7	–
UA-CMDet [23]	RGB + T	87.5	60.7	46.8	87.1	38.0	64.0	13.2
Halfway Fusion [25]		90.1	62.3	58.5	89.1	49.8	70.0	29.0
CIAN [28]		90.1	63.8	60.7	89.1	50.3	70.8	31.4
AR-CNN [55]		90.1	64.8	62.1	89.4	51.5	71.6	26.5
MBNet [30]		90.1	64.4	62.4	88.8	53.6	71.9	32.3
TSFADet [46]		89.9	67.9	63.7	89.8	54.0	73.1	28.0
C ² Former [31]		90.2	68.3	64.4	89.8	58.5	74.2	–
OAFa [20]		90.3	76.8	73.3	90.3	66.0	79.4	55.1
RFVTD (Ours)		89.6	77.1	73.8	90.4	70.9	80.5	50.8

Decomposition Module (SPDM, A1), the Adaptive Modality Semantic Association and Enhancement Module (AMSEM, A2), and the Dual-Supervised Feature Alignment module (DSFA, A3). We further examine their pairwise combinations (A4–A6) and the full integration of all three modules (A7).

As shown in Table 6, each proposed module improves over the YOLOv5s + Add baseline. Among the single-module variants, DSFA brings the largest gain, increasing mAP from 45.1% to 53.6%. This indicates that progressive cross-modal alignment is critical for detecting UAVs from misaligned RGB-T inputs. SPDM and AMSEM also provide clear improvements, reaching 48.6% and 48.7%, respectively. These results show the benefits of separating modality-invariant and modality-specific cues, as well as introducing coarse semantic association and target-aware enhancement.

Combining the modules further improves performance. SPDM + AMSEM reaches 49.3%, suggesting that feature decomposition facilitates cross-modal association. With DSFA, SPDM and AMSEM further improve performance to 54.6% and 54.4%, respectively, indicating that residual alignment benefits from disentangled representations and coarse association priors. The full model achieves the best result of 56.1%, improving the baseline by 11.0 points. These results show that SPDM, AMSEM, and DSFA are complementary for registration-free RGB-T UAV detection.

4.6.2. Q2. SPDM constraints

To further examine the three constraints in SPDM, we conduct a constraint-level ablation study on the Anti-UAV dataset. As shown in Table 7, the full model achieves the best performance of 56.1% mAP. Among the individual constraints, removing the reconstruction con-

Table 5
Performance comparison on the LLVIP dataset under well-aligned low-light conditions.

Methods	Modality	mAP (%) $\uparrow$
Faster R-CNN [48]	RGB	89.0
YOLOv5s [47]		90.8
Faster R-CNN [48]	T	93.7
YOLOv5s [47]		94.6
UA-CMDet [23]	RGB + T	96.3
MBNet [30]		93.2
CMA-Det [22]		97.1
RSDet [52]		95.8
RFVTD (Ours)		96.9

Table 6
Ablation study on module contributions (Q1) using the Anti-UAV dataset.

Method	SPDM	AMSEM	DSFA	mAP (%)
A0: Baseline (YOLOv5s + Add)				45.1
A1: + SPDM	✓			48.6
A2: + AMSEM		✓		48.7
A3: + DSFA			✓	53.6
A4: SPDM + AMSEM	✓	✓		49.3
A5: SPDM + DSFA	✓		✓	54.6
A6: AMSEM + DSFA		✓	✓	54.4
A7: Full (SPDM + AMSEM + DSFA)	✓	✓	✓	56.1

Table 7
Ablation study of individual constraints in SPDM on the Anti-UAV dataset. Values in parentheses indicate the mAP change relative to the full model.

Variant	$\mathcal{L}_{rec}$	$\mathcal{L}_{\perp}$	$\mathcal{L}_{dom}$	mAP (%) $\uparrow$
Full	✓	✓	✓	56.1
w/o reconstruction		✓	✓	54.8 (−1.3)
w/o decorrelation	✓		✓	55.5 (−0.6)
w/o adversarial constraint	✓	✓		55.7 (−0.4)
w/o all SPDM constraints				54.2 (−1.9)

straint causes the largest performance drop of 1.3 mAP points, indicating that information-preserving reconstruction is important for preventing degenerate shared-private decomposition. Removing the decorrelation and adversarial constraints leads to decreases of 0.6 and 0.4 points, respectively, showing that reducing shared-private redundancy and promoting modality invariance in the shared representation both benefit subsequent cross-modal association and alignment. When all three constraint losses are removed while retaining the shared-private projection architecture, mAP drops by 1.9 points, confirming that the gain of SPDM comes not merely from the additional branches, but also from the proposed decomposition constraints.

4.6.3. Q3. Alignment direction

To examine the effect of alignment direction in DSFA, we compare three variants on the Anti-UAV dataset: uni-directional alignment from RGB to T (R→T), uni-directional alignment from T to RGB (T→R), and bidirectional alignment. The baseline without DSFA is also reported for reference. Table 8 shows that introducing DSFA alignment markedly improves accuracy over the non-alignment baseline (49.3%). Both uni-directional settings achieve notable gains, with R→T reaching 54.4% and T→R slightly higher at 54.7%. However, either uni-directional setting leaves the opposite branch unaligned and cannot fully resolve cross-modal discrepancies. By contrast, the bidirectional variant achieves 56.1%, outperforming both uni-directional designs and confirming that symmetric alignment better handles modality differences under misaligned inputs.

4.6.4. Q4. Number of refinement stages K

To investigate the effect of progressive alignment depth, we vary the number of refinement stages K in DSFA. Specifically, we evaluate

Table 8
Ablation study of DSFA alignment direction (Q3) on the Anti-UAV dataset.

Method	Uni (R→T)	Uni (T→R)	mAP (%)
B0: w/o DSFA			49.3
B1: Uni (R→T)	✓		54.4
B2: Uni (T→R)		✓	54.7
B3: Bidirectional	✓	✓	56.1

$K = \{0, 1, 2, 3, 4, 5\}$. Fig. 6 shows that increasing K generally improves detection performance compared with shallow or non-refined alignment. The most notable gains appear on Anti-UAV and DVTOD, confirming that progressive residual refinement is particularly beneficial under severe modality misalignment. DroneVehicle shows moderate improvements, while LLVIP remains relatively saturated due to its well-aligned inputs. Although some datasets obtain marginally higher mAP when using more than three refinement stages, the improvement beyond $K = 3$ is limited and comes with additional computational cost. In particular, deeper refinement may also increase the risk of over-warping when the modalities are already well aligned. Therefore, we adopt $K = 3$ as the default setting, since it provides a favorable trade-off between accuracy, stability, and efficiency across datasets.

4.7. Qualitative analysis of feature alignment

Fig. 5 visualizes the T-to-R direction as a representative example of DSFA feature alignment under different cross-modal offsets. Before alignment, the thermal shared-feature responses are often displaced from the corresponding RGB targets. After alignment, the dominant responses shift toward the RGB target regions and become more concentrated, showing that DSFA effectively corrects cross-modal spatial misalignment.

The last two examples further show that, when target features are weak and background interference is strong, the unaligned features can produce misleading off-target responses. DSFA reduces these responses and redirects the activations toward the targets, improving the spatial consistency of cross-modal target features.

4.8. Evaluation on real-world data

4.8.1. Data collection and annotation

We further evaluate RFVTD on 1,000 real-world RGB–T image pairs collected with an HP-DMS15 electro-optical platform. The RGB camera uses a 1/2.8-inch CMOS sensor with a resolution of 1920×1080 , while the thermal camera uses an uncooled VOx detector with a resolution of 640×512 and a spectral range of 8–14 μm . The two sensors are rigidly mounted and electronically synchronized for frame-level temporal correspondence. DJI Matrice 350 RTK and DJI Matrice 4E are used as target UAVs. Both modalities are manually annotated with instance-level bounding boxes. RGB annotations are first generated and then refined using thermal observations and, when needed, temporal motion cues. No pixel-level registration is applied to the RGB–T pairs. The 1,000 pairs are evenly split into 500 pairs for fine-tuning and 500 pairs for testing. All compared methods are fine-tuned on the same 500 pairs and evaluated on the same held-out test set.

Background interference is divided into three levels according to scene complexity. Samples are first grouped based on target-background texture similarity and then manually refined according to clutter density and target-background similarity. No interference denotes a clearly distinguishable UAV against a relatively clean background. Weak interference includes moderate target-background similarity or sparse clutter, while strong interference corresponds to high similarity and dense clutter that substantially increase detection difficulty. The dataset contains 200, 300, and 500 pairs with no, weak, and

Table 9

Detection performance on the collected real-world RGB-T dataset under different background interference levels.

Scene	Methods	mAP (%) ↑
No interference	YOLOv5s + Add	79.3
	YOLOv5s + CMX	82.5
	CMA-Det	84.1
	RFVTD (Ours)	85.5
Weak interference	YOLOv5s + Add	64.3
	YOLOv5s + CMX	68.8
	CMA-Det	70.6
	RFVTD (Ours)	74.5
Strong interference	YOLOv5s + Add	29.5
	YOLOv5s + CMX	33.4
	CMA-Det	39.7
	RFVTD (Ours)	46.4

Table 10

Accuracy and computational complexity comparison on the Anti-UAV detection split. FLOPs are measured at 640×640 input resolution, and FPS is measured with a batch size of 1 on a single NVIDIA A100 GPU.

Method	mAP(%) ↑	Params(M) ↓	FLOPs(G) ↓	FPS ↑
YOLOv5s + Add	45.1	13.5	33.2	63.7
CMA-Det	53.9	33.7	36.7	55.6
RFVTD (Ours)	56.1	22.6	37.4	50.8

strong interference, respectively. Representative examples are shown in Fig. 7.

4.8.2. Evaluation under background interference

Table 9 reports the results on the held-out test set under different background-interference levels. RFVTD achieves the best performance in all three conditions, with mAP scores of 85.5%, 74.5%, and 46.4% under no, weak, and strong interference, respectively. Compared with CMA-Det, RFVTD gains 1.4, 3.9, and 6.7 percentage points, with larger improvements as interference increases.

All methods degrade substantially under strong interference, where dense clutter and visually or thermally similar distractors make small UAVs difficult to distinguish. RFVTD still performs best, but its mAP drops to 46.4%, indicating that highly complex scenes remain challenging. This drop may also relate to the limited coverage of strong-interference cases in the Anti-UAV training data, which is only partly addressed by the real-world fine-tuning set. Future work will focus on constructing more challenging UAV datasets and improving robustness under strong background interference.

4.9. Computational complexity

Table 10 compares the accuracy and computational complexity of RFVTD with representative RGB-T baselines. All methods are evaluated under the same inference settings. FLOPs are measured at 640×640 input resolution, and FPS is tested with a batch size of 1 on a single NVIDIA A100 GPU. RFVTD achieves the highest mAP of 56.1% with 22.6M parameters and 37.4G FLOPs. Compared with CMA-Det, it improves mAP by 2.2 points with 32.9% fewer parameters and only 1.9% more FLOPs. RFVTD also maintains an inference speed of 50.8 FPS, demonstrating a favorable accuracy–efficiency trade-off.

5. Conclusion

In this paper, we proposed RFVTD, a registration-free visible–thermal fusion framework for UAV detection. RFVTD directly processes unregistered RGB-T image pairs without requiring image-level pre-registration or pixel-level alignment supervision. The framework

combines SPDM for shared–private feature decomposition, AMSEM for coarse cross-modal semantic association and initial deformation prior estimation, and DSFA for progressive residual feature alignment. Through this design, RFVTD separates modality-invariant structural cues from modality-specific details, establishes non-local cross-modal correspondence, and refines feature-level misalignment in a detection-driven manner. Extensive experiments on Anti-UAV, DroneVehicle, DV-TOD, and LLVIP demonstrate that RFVTD is particularly effective under misaligned conditions and remains competitive on well-aligned data. Ablation studies verify the complementary roles of SPDM, AMSEM, and DSFA, while real-world evaluations further show the robustness of RFVTD against complex background interference. Nevertheless, extremely cluttered scenes and strong distractors can still degrade detection reliability. Future work will focus on constructing more challenging UAV datasets, improving robustness under severe background interference, and exploiting temporal consistency in video-based UAV detection.

CRediT authorship contribution statement

Shenghui Huang: Writing – original draft, Validation, Software, Methodology, Formal analysis, Conceptualization; **Yuqi Wang:** Validation, Software, Methodology, Formal analysis; **Longkun Zou:** Validation, Software, Methodology, Formal analysis; **Lingwei Dang:** Validation, Investigation, Data curation; **Hongyu Chi:** Validation, Investigation, Data curation; **Xiaopei Chen:** Writing – review & editing, Visualization, Investigation, Data curation; **Ke Chen:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Conceptualization; **Qingyao Wu:** Writing – review & editing, Supervision, Project administration, Funding acquisition.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used ChatGPT (OpenAI) to improve language and readability. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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